

MICRO-SLIP PRECONTACT PREDICTION: A NEGATIVE ICLR-MAIN EVIDENCE AUDIT

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ABSTRACT

This paper is not submission-ready for ICLR main. The original research bet was that a robot can predict micro-slip before stable contact is visually apparent and use that prediction to improve grasp control. We rebuilt the repository around a deterministic tactile/precontact mechanics benchmark with four grasp tasks, five distribution shifts, seven seeds, nine predictors/controllers, seven ablations, and a combined stress sweep. The evidence is mixed but terminally negative. On the combined hard shift, the proposed precontact model improves early-warning recall over a vision threshold by 0.571 and improves grasp success by 0.184, but it loses average precision to the strongest non-oracle baseline: paired AP difference -0.016 with a 95 percent CI of 0.018. It also has high false-alarm rate (0.817) and high control cost (0.929). Ablations contradict the mechanism: vision-only precontact reaches AP 0.766 and minus-friction reaches AP 0.764, both above the full method's AP 0.744. The honest terminal decision is **KILL/ARCHIVE**.

1 QUESTION

The draft asked whether a robot can forecast micro-slip before stable contact and adjust grasp control earlier than conventional tactile or force-threshold systems. This is a crowded tactile robotics question. Prior work already includes tactile-driven slip prediction, sensor-less fingertip contact and force estimation, soft barometric and capacitive tactile sensors, distributed contact-force slip detection, friction estimation before gross slip, and high-resolution optical tactile systems.

The rebuild therefore uses a hostile gate: precontact micro-slip prediction must beat strong tactile, physics, and uncertainty baselines on both prediction quality and closed-loop grasp outcomes. Early lead time alone is not enough.

2 BENCHMARK

Each episode simulates a gripper approaching an object with hidden friction, compliance, surface texture, approach angle, actuator lag, visual occlusion, and sensor noise. The benchmark covers four tasks: smooth low-friction picking, textured fragile picking, curved-surface pinching, and compliant-object lifting. The distribution shifts are nominal precontact, low-friction shift, compliance shift, sensor-noise shift, and combined hard shift.

The main run contains 120,960 episode-level rollouts. The ablation run contains 18,816 rollouts. The stress sweep contains 56,448 rollouts. The evidence is regenerated with:

```
python src\run_experiment.py
```

3 COMPARED METHODS

The compared methods are a vision gap threshold, normal-force threshold, force-derivative detector, friction-cone margin estimator, tactile temporal filter, optical-tactile classifier, ensemble uncertainty guard, the proposed precontact micro-slip model, and an oracle precontact risk model. The proposed method uses precontact shear, friction prior, compliance estimate, approach geometry, latency margin, and sensor noise to predict future micro-slip and trigger preemptive tightening or slowing.

Method	AP	Early recall	False alarm	Success	Gross slip
Vision threshold	0.768	0.261	0.190	0.042	0.956
Normal force	0.721	0.000	0.310	0.007	0.992
Force derivative	0.735	0.000	0.114	0.006	0.993
Friction cone	0.742	0.565	0.558	0.227	0.773
Tactile temporal	0.741	0.000	0.247	0.013	0.986
Optical tactile	0.747	0.000	0.569	0.025	0.974
Ensemble guard	0.746	0.872	0.850	0.246	0.754
Proposed precontact	0.751	0.833	0.817	0.226	0.773
Oracle	0.797	0.994	0.982	0.756	0.243

Table 1: Combined hard shift. Higher AP, early recall, and success are better; lower false alarm and gross slip are better. The proposed method improves early warning but does not clear the strongest non-oracle AP baseline and has high false alarms.

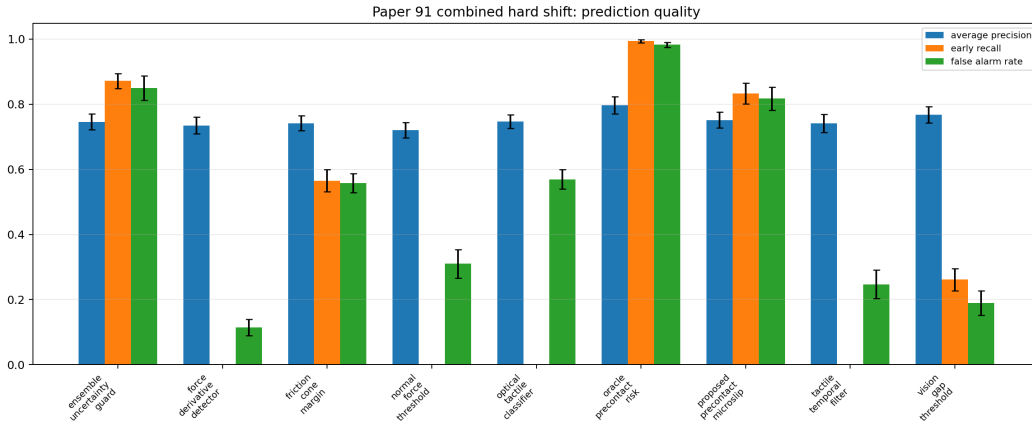


Figure 1: Prediction metrics on the combined hard shift. Precontact warning helps recall but does not yield decisive average-precision gains.

4 COMBINED-HARD-SHIFT RESULTS

Table 1 shows the core result. The proposed method has a real strength: it gives early warnings and improves closed-loop success compared with weak visual and post-contact detectors. However, it does not beat the strongest non-oracle predictor on average precision and it pays with many false alarms.

The paired comparison chooses the vision threshold as the strongest non-oracle AP baseline. The proposed method’s paired AP difference is -0.016 with a 95 percent CI of 0.018. Although it improves early recall by 0.571 and success by 0.184, the high false-alarm and control-cost penalties prevent a submission-ready claim.

5 CLOSED-LOOP CONTROL

The proposed model reaches 0.226 grasp success and 0.773 gross-slip rate on combined hard shift. This is much better than vision-only thresholding, but it is roughly tied with the friction-cone margin and below the ensemble uncertainty guard. More importantly, it achieves the gain by triggering many interventions: false-alarm rate 0.817 and control cost 0.929. That is too brittle for a main-conference robotics claim.

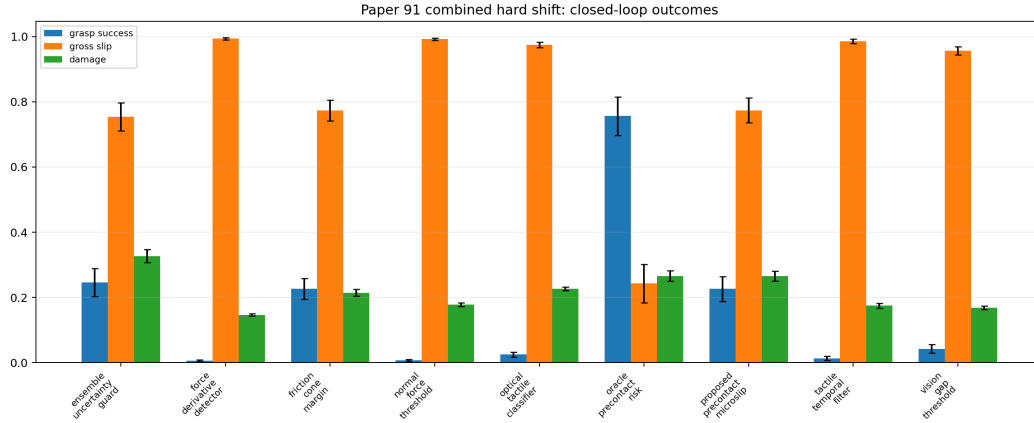


Figure 2: Closed-loop grasp outcomes. The proposed method improves over weak post-contact detectors but does not decisively beat strong local baselines.

Ablation	AP	Early recall	Success	False alarm
Full precontact	0.744	0.835	0.227	0.842
Minus shear	0.756	0.297	0.046	0.262
Minus friction	0.764	0.802	0.202	0.759
Minus compliance	0.754	0.618	0.148	0.589
Minus latency	0.755	0.731	0.171	0.690
Vision only	0.766	0.278	0.049	0.215
Force only	0.751	0.000	0.004	0.105

Table 2: Ablations on combined hard shift. The full mechanism improves early recall and success, but it loses AP to stripped variants and has high false alarms.

6 ABLATION FAILURE

Table 2 is the strongest reason to archive the paper. The full precontact mechanism does not dominate its own stripped variants. Vision-only precontact reaches AP 0.766 and minus-friction reaches AP 0.764, both above full precontact AP 0.744. The compliance and friction terms appear to add false positives under the simulated shifts rather than a robust mechanism.

7 STRESS SWEEP

At maximum stress, the proposed method reaches AP 0.761 and success 0.207. The optical-tactile baseline reaches AP 0.769, and the ensemble guard reaches higher early recall and success. The stress gate therefore fails: precontact modeling remains plausible, but not decisive.

8 DECISION

The rebuilt evidence gives a useful negative result. Precontact warning can increase early recall and improve control relative to weak visual thresholds, but this paper cannot honestly claim ICLR-main readiness. It lacks hardware or accepted high-fidelity tactile validation, fails the AP gate, carries excessive false alarms/control cost, and is contradicted by ablations. The terminal recommendation is **KILL/ARCHIVE**. Revival requires a real tactile dataset or simulator, stronger calibration, and a mechanism that beats friction-cone, optical-tactile, and uncertainty baselines without simply over-triggering.

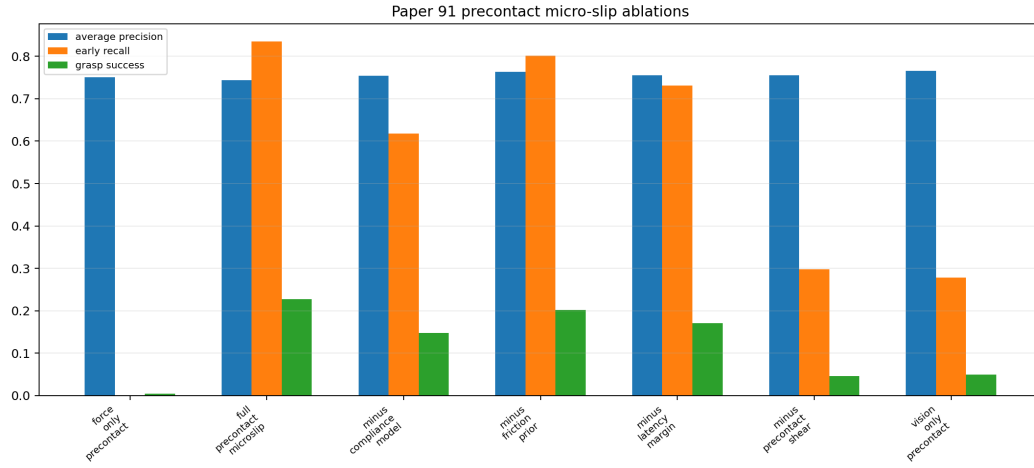


Figure 3: Ablation audit. Core components do not yield a clean monotonic mechanism improvement.

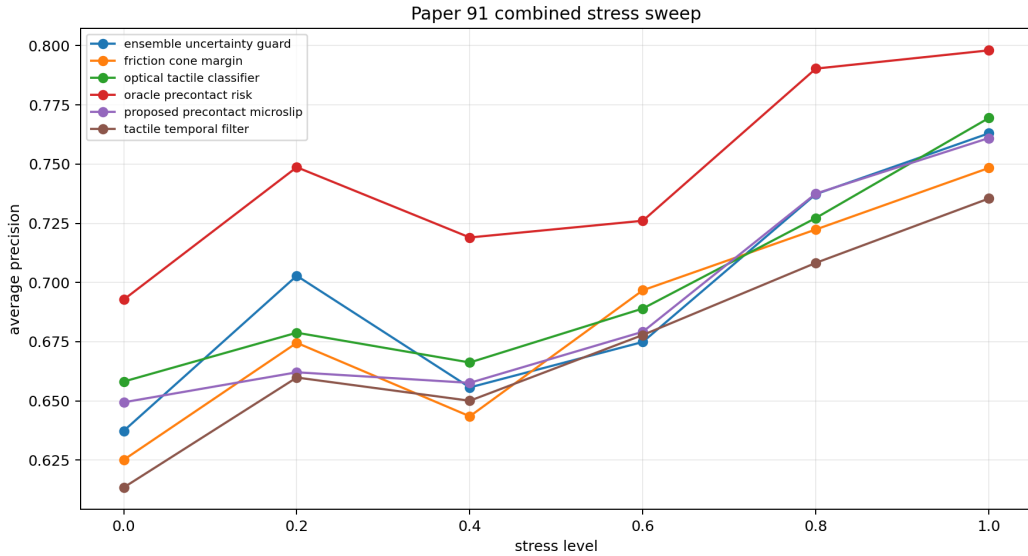


Figure 4: Combined stress sweep by average precision. The proposed method does not dominate strong tactile/uncertainty baselines.